

(continued)

```
$ complete -f -d -W "runserver createsuperuser test shell dbshell \
migrate makemigrations loaddata dumpdata" ./manage.py
```

This allows you to type `manage.py`, the first letter of a management command, and then press `TAB` to auto-complete the name of the command, saving even more typing. Other shells have equivalent completion mechanisms; in fact, some versions of `zsh` ship with auto-completion for the Django command built-in.

Once going, the development server shows you some debug info and then tells you its base URL. By default, the server listens on port 8000 of your local machine. If you prefer a different port number, you can provide it as part of the `runserver` command (e.g., `python manage.py runserver 8800`). The output from the server command looks like this:

```
Watching for file changes with StatReloader
Performing system checks...
```

```
System check identified no issues (0 silenced).
```

```
You have 18 unapplied migration(s). Your project may not work properly
until you apply the migrations for app(s): admin, auth, contenttypes,
sessions. Run 'python manage.py migrate' to apply them.
```

```
Django version 5.0.2, using settings 'RiffMates.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CONTROL-C.
```

A warning that can
be ignored for now

Listening on
port 8000

Don't worry about the 18 unapplied migration(s) message—you'll deal with it shortly. Remember that `localhost` is an alias to the IP address `127.0.0.1`, so you can also use the English alias `http://localhost:8000/` to visit your site. Enter the base URL into your browser and see the rocket ship on the default Django project page (figure 2.2).

Dev servers are for development

For convenience, Django includes a web server. This means you don't need to install and manage one when you're creating and maintaining your website. This is not a hardened web server though, and it should never be put into an untrusted environment.

Do not expose the Django development server directly to the internet, and if you're using it internally at a company, speak with your IT department about whether it should be hosted behind your firewall. Furthermore, the default configuration for a Django project is in debug mode, which also should not be used in production. Appendix B discusses what you need to know to put your Django project into a hardened production environment. It is right there in the name: the *development* server should only be used for *development*!